

# Emerging Respiratory Virus SPOK-24 (Formerly Virus X) – Australia

## Situation Report – 1, published 25 October 2024

### Highlights

- As of October 20, a total of 85 confirmed hospitalizations across multiple Australian cities for severe respiratory infections negative for common respiratory viruses.
- Seven confirmed deaths, including two pediatric fatalities, all testing positive for Virus SPOK-24.
- Clear evidence of sustained human-to-human transmission through household and nosocomial spread.
- Virus X genome publicly available on GenBank; PCR primers developed by the Doherty Institute on October 22.
- WHO is increasing its risk assessment and preparing for potential further global spread.

| CURRENT EPIDEMIOLOGICAL SITUATION                                                                                                           |              |                         |
|---------------------------------------------------------------------------------------------------------------------------------------------|--------------|-------------------------|
| 1. Total Confirmed Hospitalizations (Negative for Common Respiratory Panels, Under Investigation for Virus SPOK-24) as of October 21, 2024. |              |                         |
| Location                                                                                                                                    | Total Cases  | New Cases (Past 5 Days) |
| Brisbane Metro, QLD                                                                                                                         | 75           | +18                     |
| Townsville, QLD                                                                                                                             | 1            | +0                      |
| Rockhampton, QLD                                                                                                                            | 1            | +1                      |
| Sydney, NSW                                                                                                                                 | 8            | +5                      |
| TOTAL                                                                                                                                       | 85           | +24                     |
| 2. Confirmed Deaths (All Virus SPOK-24 Positive) as of October 21, 2025                                                                     |              |                         |
| Location                                                                                                                                    | Total Deaths | New Deaths              |
| Brisbane Metro, QLD                                                                                                                         | 6            | +3                      |
| Townsville, QLD                                                                                                                             | 1            | +1                      |
| TOTAL                                                                                                                                       | 7            | +4                      |

### 3. Transmission Patterns

- Multiple confirmed household transmission events indicate sustained human-to-human spread.
- Nosocomial (hospital-acquired) transmission has been observed in at least three Brisbane hospitals.
- No confirmed international cases at this stage, but WHO recommends global surveillance.

## Public Health Actions Taken

- Testing & Diagnostics: All clinical samples from hospitalized respiratory patients negative for common viruses are now being tested for Virus SPOK-24 at the Doherty Institute. PCR primers for Virus SPOK-24 have been developed.
- Case Identification & Contact Tracing: Expanded hospital surveillance and active contact tracing are in place.
- Hospital Preparedness & Response: Infection control measures strengthened in hospitals with confirmed cases.
- WHO Global Actions: WHO is assessing the risk of international spread and coordinating with Australian health authorities.

## Risk Assessment

- Current Global Public Health Risk: HIGH (Pending Further Investigations).
- Likelihood of Global Spread: MODERATE-HIGH (Based on sustained transmission and domestic spread to multiple cities).
- Risk of Severe Disease: Elevated in young children and elderly populations.

## Recommendations For WHO Member States

- Enhanced Surveillance: Countries should increase screening for severe acute respiratory infections (SARI) and atypical pneumonia cases.
- Strengthened Laboratory Testing & Sequencing: National labs should obtain Virus SPOK-24 PCR primers and sequence positive cases where possible.
- Healthcare Preparedness: Hospitals should be on alert for unexplained severe respiratory cases.
- Public Health Communications: Governments should regularly update the public and healthcare workers on the evolving situation.

## Next Steps & Global Response

- WHO Western Pacific Regional Office (WPRO) continues close collaboration with Australian health authorities.
- An Emergency Committee meeting is under consideration to assess whether this constitutes a Public Health Emergency of International Concern (PHEIC).
- WHO will issue an updated Situation Report in 48 hours or as more information becomes available.

*For urgent inquiries and international reporting, contact the WHO IHR Emergency Team.*

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